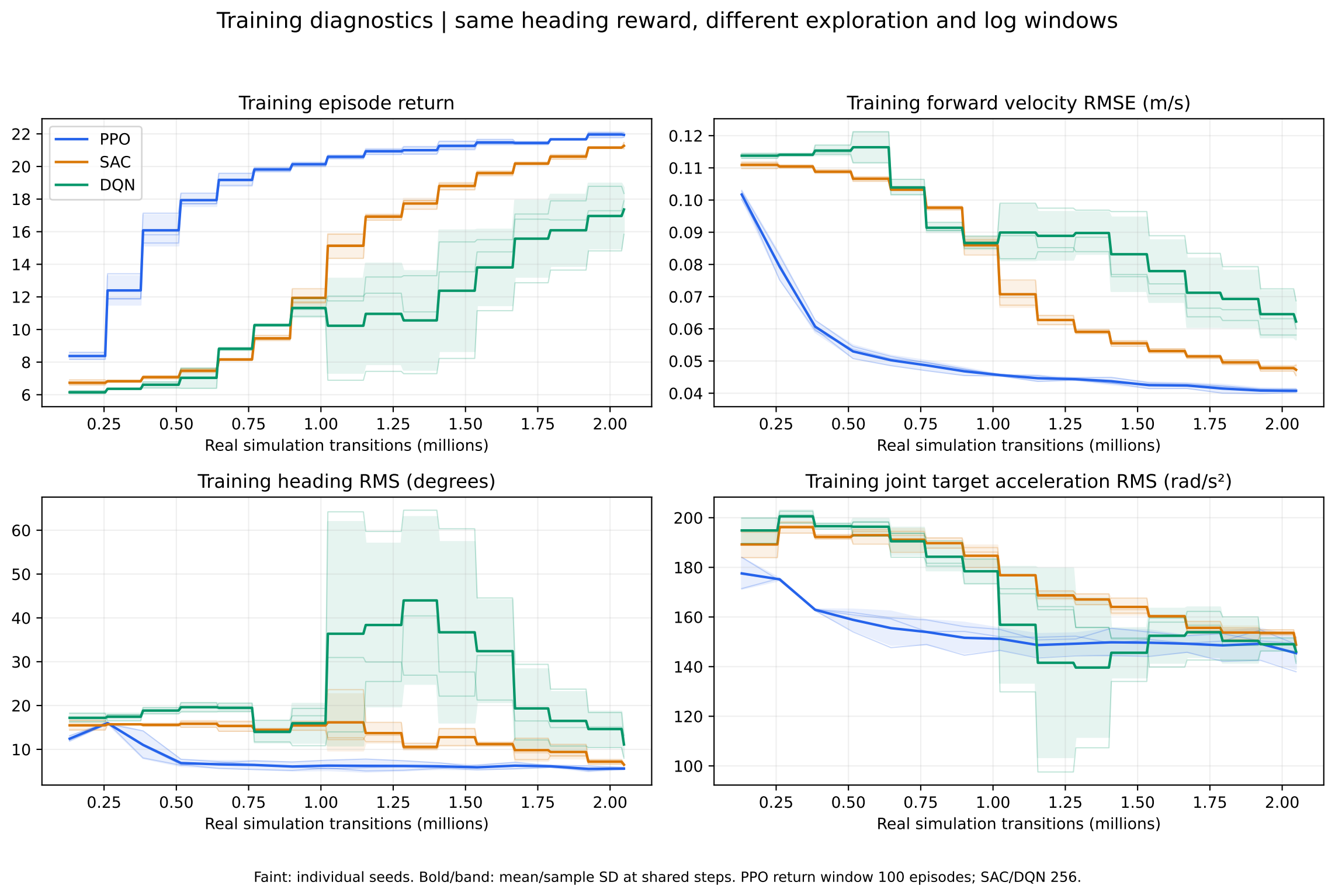
Training diagnostics | same heading reward, different exploration and log windows
Training episode return
PPO
SAC
DQN
Real simulation transitions (millions)
Training forward velocity RMSE (m/s)
Real simulation transitions (millions)
Training heading RMS (degrees)
Real simulation transitions (millions)
Training joint target acceleration RMS (rad/s²)
Real simulation transitions (millions)
Faint: individual seeds. Bold/band: mean/sample SD at shared steps. PPO return window 100 episodes; SAC/DQN 256.